Course Syllabus

EE 582: Advanced Topics (Energy Systems Modeling and Simulation)

Fall 2025

Credits: 3

Instructor: Dr. Seyyedmilad (Milad) Ebrahimi Email: s.ebrahimi@wsu.edu

Lectures: **Tuesdays** and **Thursdays** 4:20 – 5:35 pm

Location: See below

Pullman	Murrow Hall 229	In Person
Vancouver	Engineering and CS Bldg 209	Video Conference via Zoom
Tri-Cities	Consol Info Ctr 221	Video Conference via Zoom

Attendance: Students are expected to attend all the lectures. For students in Pullman attendance is in-person, while those in other campuses should use the video-conference platform (Zoom) and have their video ON during the lecture and participate in discussions. **Any absence should be communicated with the instructor in-advance with a legitimate reason.** More than 3 unexcused absences can result in loss of one letter grade.

Office Hours: Email the instructor for an appointment with "EE 582" in the subject line.

Class Notes: Instructor's class notes will be posted (in PDF format) on Canvas before each lecture. Students are encouraged to take their own notes during the lectures and participate in class discussions. Lectures will also be video-captured (automatically through Zoom) and will be available in Zoom cloud.

Assignments: There will be **7–8** assignments throughout the course. Assignment questions will be posted on Canvas. The submission link will be open until **11:59 pm** of the due date (the due date is normally in the title of the assignment file). Late submissions will result in reduced mark (10% per day, up to 3 days).

Simulation Project: Students need to select one IEEE Transaction paper related to the course topics (by the end of October 2025) and implement the models/methods and replicate the results (the authors of the papers cannot be affiliated with WSU or student's previous affiliation). A written report should be submitted (together with the simulation files) and also presented orally no later than the last day of classes.

Final Evaluation: The final grade will be based on the following weighting scheme:

Assignments (80%) – Project (20%)

The corresponding grade letters are based on the following scheme:

Score	Letter Grade
90 – 100	A
85 – 89	A–
80 – 84	B+
75 – 79	B
70 – 74	B–
below 70	Aim that you should never be here!

Accommodations: Reasonable accommodations are available for students with disability. Accommodations should be documented and approved through the Access Center. You can visit or call the Access Center (Washington Building 217; (509) 335-3417; Access.Center@wsu.edu) to schedule an appointment with an Access Advisor. For more information, visit http://accesscenter.wsu.edu.

Academic Integrity: The academic enterprise is founded on the basis of honesty, civility, and integrity. This also means you should not cheat, copy, or mislead others about your work. Students are encouraged to discuss among themselves the problems in each assignment. However, the turned-in assignments and project must show the individual work and reflect the individual understanding of the material by each student. **Reports suspected of cheating will not be graded (zero mark!), and may qualify for withdrawal from the course and/or suspension from the University.** Please review the Washington State University Standards of Conduct for Students here. All instances of cheating will be reported according to this policy.

Tentative Outline:

1. **Module 1:** Introduction to energy conversion systems, applications of power-electronic converters in power systems, and electric machinery drives
2. **Module 2:** Introduction to software tools and methods for electromagnetic transient (EMT) simulations
3. **Module 3:** Nodal-analysis-based methods for EMT simulations of power systems, numerical integration methods, accuracy, and stability
4. **Module 4:** State-variable-based methods for EMT simulations, numerical methods, stiffness and stability, variable-step ODE solvers and step-size control, and multi-step methods
5. **Module 5:** Space phasors and modeling of power systems in two-dimensional reference frames
6. **Module 6:** Control and modeling of two-level and three-level voltage source converters (VSCs) including their average-value models
7. **Module 7:** EMT modeling and control of synchronous machines
8. **Module 8:** EMT modeling and control of induction machines
9. **Module 9:** Grid-following and grid-forming converters and their control
10. **Module 10:** Introduction to real-time simulations and constraints

The above tentative topics are covered over the **16 Weeks** of the semester with the following tentative plan:

Month	Week	Date	Tentative Topic
Aug.	1	19	Module 1
		21	Students' Introduction
	2	26	Module 2
		28	Module 3
Sep.	3	2	Module 3
		4	Module 3
	4	9	Module 3
		11	Module 3
	5	16	Module 3
		18	Module 4
	6	23	Module 4
		25	Module 4
	7	30	Module 4
Oct.		2	Module 4
	8	7	Module 4
		9	Module 5
	9	14	Module 6
		16	Module 6
	10	21	Module 7
		23	Module 7
	11	28	Module 7
		30	Module 8
	12	4	Module 8
		6	Module 9
Nov.	13	11	Veterans Day – Holiday
		13	Module 9
	14	18	Module 9
		20	Module 9
	15	25	Thanksgiving Vacation
		27	
Dec.	16	2	Module 10
		4	Module 10

Reference Books:

- *EMTP Theory Book* by Hermann W. Dommel
- *Voltage-Sourced Converters in Power Systems* by Amirnaser Yazdani, Reza Iravani
- *Analysis of Electric Machinery and Drive Systems* by P.C. Krause
- *Real-Time Simulation Technology for Modern Power Electronics* by H. Bai, C. Liu

Disclaimer: Information contained in this document may change as required during the semester and will be communicated to students via email, in class, and/or on Canvas. Students should attend all lectures to stay up-to-date.

University Syllabus

Students are responsible for reading and understanding all university-wide policies and resources pertaining to all courses (for instance: accommodations, crisis resources, policies on discrimination or harassment), which can be found in the university syllabus:

https://syllabus.wsu.edu/university-syllabus